

SC224: Probability and Statistics

Course Placement: Probability and Statistics is a core course for second year students of BTech ICT program.

Course Format: It is **3 hours' lecture and 1-hour tutorial** every week.

Course Contents:

Probability: Sample Space, Events, Axioms of Probability, Conditional Probabilities, Independence of Events, Bayes' Theorem, Random Variables, Types of Random Variables, Jointly Distributed Random Variables, Expectation, Variance, Sums of Random Variables, Covariance, Markov's and Chebyshev's Inequality, Law of Large numbers, Special Random Variables: Binomial, Poisson, Geometric, Uniform, Normal, Exponential, Chi-Square, t-Distribution, Sample Mean, Central Limit Theorem, Sample Variance.

Statistics: Parameter Estimation: maximum likelihood estimators, interval estimates, Hypothesis Testing: Significance Level, Mean and Variance of Normal Population, Hypothesis Testing: Mean of Poisson Distribution, Hypothesis testing in Bernoulli populations, Regression, least square estimators, distribution of estimators, Goodness of fit tests.

Text and Reference Books:

1. Introductory Statistics, Sheldon Ross, Elsevier
2. Probability and Statistics for Engineers and Scientists by Sheldon Ross, Fourth Edition

Assessment Method/Grading Policy: Two MID SEM + END SEM

Course Outcomes: This course aims at building an advanced understanding of probabilistic and statistical analysis mostly used in varied applications in engineering and science like weather prediction, complex networks, disease modeling. After completion of the course the student will have the ability to

- Apply key concepts of probability, including discrete and continuous random variables, probability distributions, conditioning, independence, expectations, variances. **P1, P2, P4**
- Define and explain the different statistical distributions (e.g., Normal, Binomial, Poisson) and the typical phenomena that each distribution often describes. **P2, P5**
- Apply the basic theorems in probability including Bayes's theorem and the CLT. **P2**
- Define and demonstrate the concepts of estimation and properties of estimators. **P4, P5**
- Apply the concepts of interval estimation and confidence intervals. **P2, P4, P5**
- Apply the concepts of hypothesis testing and p-value. **P2, P4**
- Apply the method of least squares to estimate the parameters in a regression model. **P3, P4**

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							

Lecture Schedule:

Sl. No.	Description		No. of Lectures
1	Probability		21
	1.1	Sample Space, Events, Axioms of Probability	
	1.2	Conditional Probabilities, Independence of Events, Bayes' Theorem	
	1.3	Random Variables, Types of Random Variables	
	1.4	Jointly Distributed Random Variables, Expectation, Variance	
	1.5	Sums of Random Variables, Covariance	
	1.6	Markov's and Chebyshev's Inequality, Law of Large numbers	
	1.7	Special Random Variables: Binomial, Poisson	
	1.8	Special Random Variables: Geometric, Uniform	
	1.9	Special Random Variables: Normal	
	1.10	Special Random Variables: Exponential	
	1.11	Special Random Variables: Chi-Square	
	1.12	Special Random Variables: t-Distribution	
	1.13	Sample Mean, Central Limit Theorem, Sample Variance	
2	Statistics		21
	2.1	Parameter Estimation: maximum likelihood estimators, interval estimates	
	2.2	Hypothesis Testing: Significance Level, Mean and Variance of Normal Population	
	2.3	Hypothesis Testing: Mean of Poisson Distribution	
	2.4	Hypothesis testing in Bernoulli populations	
	2.5	Regression, least square estimators, distribution of estimators	
	2.6	Goodness of fit tests	
	3.4	Channel capacity, Channel Coding Theorem, Hamming Codes	